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SAP PO Automation Simulator & Playwright RPA Engine

An enterprise-grade, end-to-end local simulation and automation system. It mimics a real SAP Fiori Purchase Order (PO) invoice verification portal and automates the multi-step verification, matching, and approval workflow using **Python, Playwright, FastAPI, Next.js 15, and SQLite**.

Technical Deep Dive: What is Happening?

The system models a real-world enterprise invoice matching workflow (Three-Way Match) where invoices are matched against purchase orders and goods receipts, audited for

price/quantity mismatches, and approved or held inside the ERP system.

1. Database Schema

(apps/shared/database.db)

- **vendors**: Stores vendor credentials, contract payment terms, address, email, phone, and GST tax registration numbers.
- **purchase_orders**: Contains purchase order headers (po_number, vendor_id, invoice_number, total amount, currency, status: **Draft** | **Pending** | **Approved** | **Rejected** | **Hold**, and created dates).
- **invoices**: Holds matching invoice metadata extracted from vendor submittals (invoice_number, PO number mapping, invoice amount, currency, status: **Pending** | **Verified** | **Approved** | **Rejected**).
- **automation_runs**: Logs every RPA subprocess transaction execution, duration speed, status (Success/Failed), timestamp, and error traces.

2. The Playwright Automation Pipeline

When you click "**Start Playwright Automation**" in the UI:

1. The **Automation Engine API (Port 8001)** receives the request and spawns the runner script (**automation_runner.py**) using the Python subprocess module.
2. Playwright launches a **Chromium browser window** on the host machine.
3. The robot signs in to the portal at **http://localhost:3000/login** using administrative credentials (**admin** / **admin123**).
4. For each purchase order listed in the uploaded Excel spreadsheet, the robot:
 - Opens the **Dashboard** and checks overall stats.
 - Navigates to the **Purchase Orders Workspace** and inputs the PO number into the search field.
 - Clicks the **Details** link to drill down into the invoice matching workspace.
 - Performs three-way verification by checking matching checklists and typing audit remarks.
 - Navigates to the **Linked Invoice** tab to audit the extracted VLM invoice scans.
 - Returns to the details tab and clicks **Approve**, **Reject**, or **Hold** depending on the instructions in the Excel sheet.
 - Saves screenshots at every major interaction checkpoint as audit logs.

5. All logs and screenshots are saved in `apps/automation-engine/outputs/{po_number}/` and can be downloaded as a single ZIP archive.

3. Critical Loop Progress Indicator

During automation execution, the Playwright RPA engine processes the list of purchase orders sequentially using a main event loop:

- **State Updates:** At the start of each iteration, the runner updates the global status file (`apps/automation-engine/outputs/status.json`) with the current PO number, index (`processed_count`), total record count (`total_count`), and success/failure counters.
- **Polling Mechanism:** The Next.js frontend polls the status API endpoint (`/api/automation-status`) every 1.2 seconds to fetch the updated progress.
- **Progress Bar:** The UI calculates the percentage completion $(\text{processed_count} / \text{total_count}) * 100$ and displays a dynamic queue progress bar, live success/failure counts, and the specific PO code currently being processed.

4. Progressive Screenshot Builds

To ensure auditability, the robot saves screenshots at 5 critical checkpoints for each PO. The frontend implements a live, progressive screenshot viewer to monitor the robot's interactions step-by-step:

- **Checkpoints:** As the runner executes the verification flow for a PO, it writes:
 1. `dashboard.png` (Portal Dashboard entry)
 2. `search-results.png` (PO Search results grid)
 3. `po-details.png` (Verification checklists and remarks input)
 4. `invoice-tab.png` (Ollama VLM OCR extraction panel)
 5. `approval-section.png` (Final status submission and toast confirmation)
- **Progressive Display:** When a PO starts, the frontend displays animated "Awaiting generation..." placeholders.
- **Live Refresh:** As the files are saved to `apps/automation-engine/outputs/{po_number}/` by the backend, the corresponding image tags resolve in the UI. The frontend appends a cache-busting timestamp query

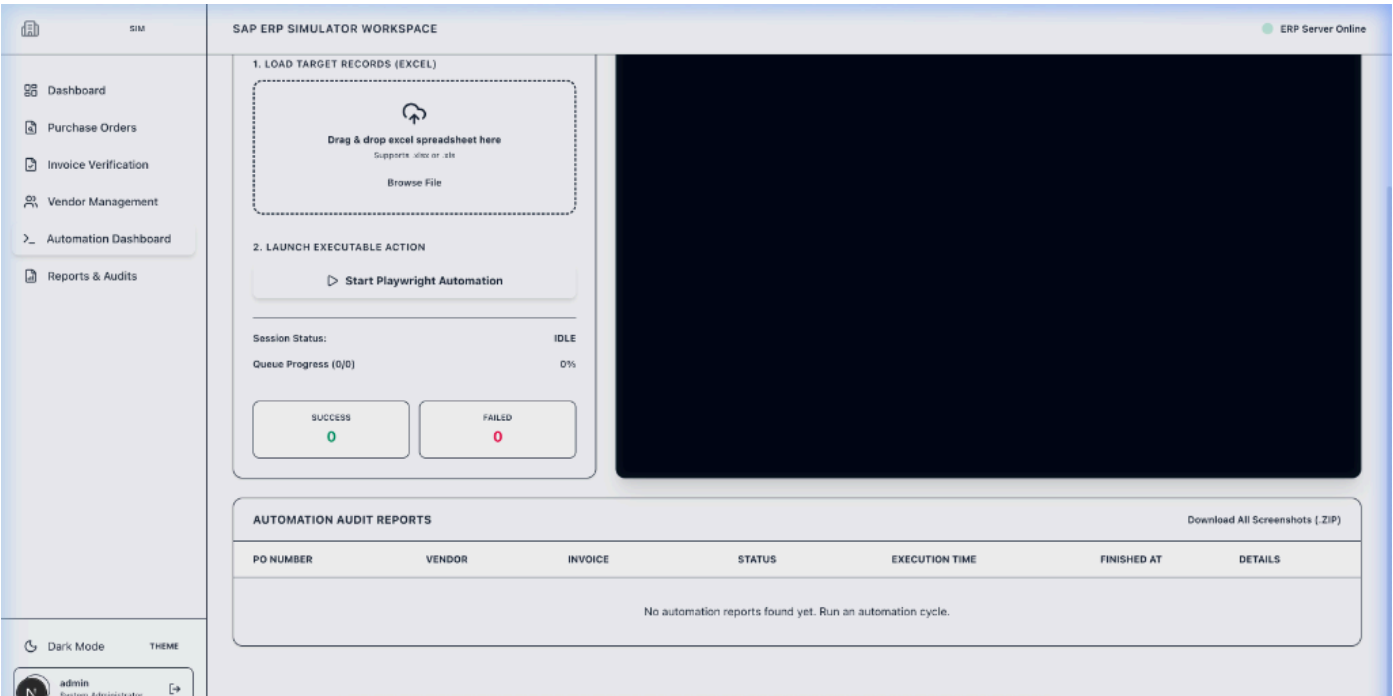
parameter (**?t=timestamp**) to force the browser to bypass local caching and pull the newly built screenshots.

RPA Automation Studio Dashboard

Below is the interface of the Next.js RPA Automation Studio frontend:

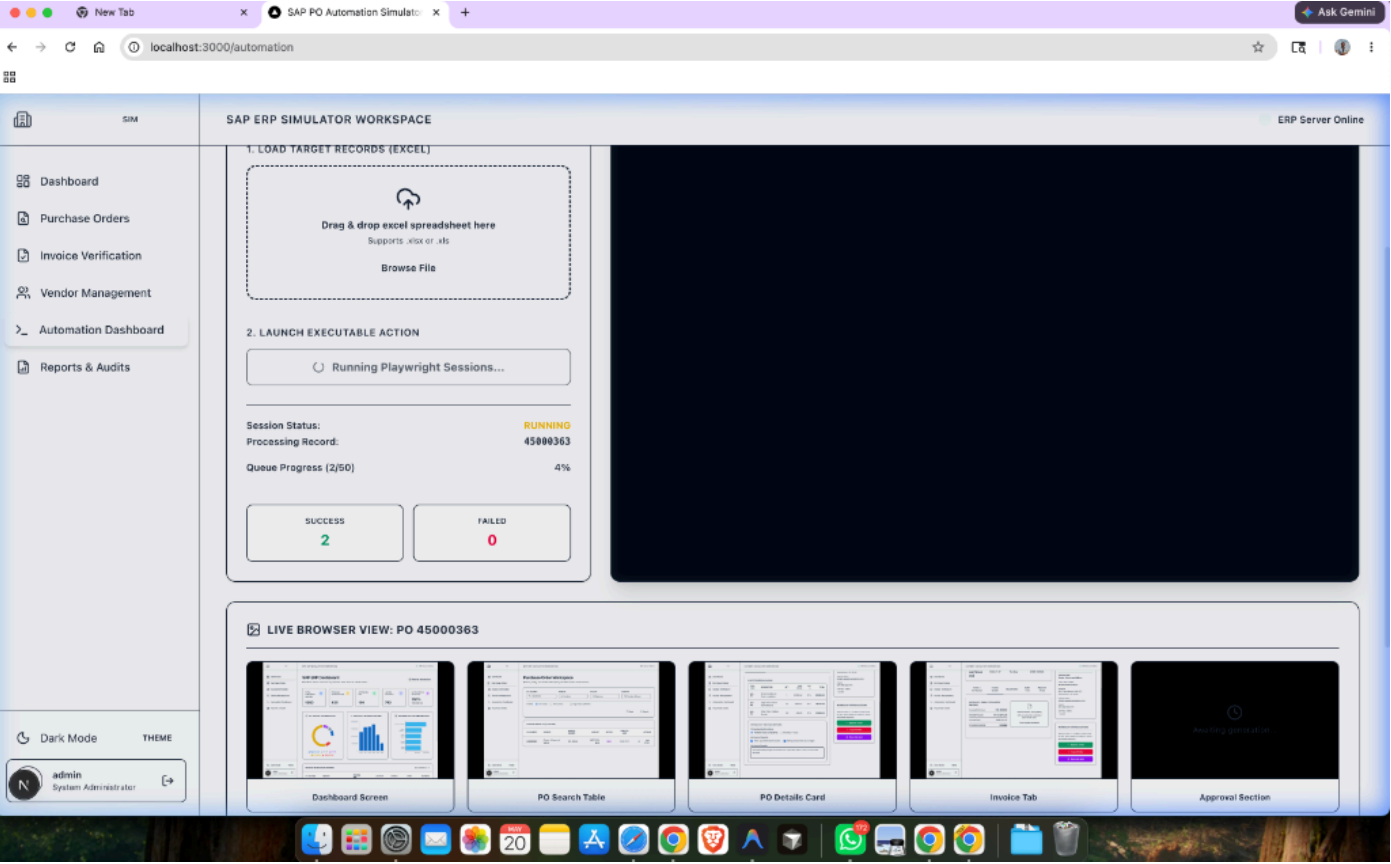
Automation Studio Dashboard (Idle)

Displays the control panel, Excel uploader, and empty audit report log before executing a run.



Automation Studio Dashboard (Running)

Shows the queue progress indicator, active stdout terminal streams, and the live, progressive screenshots showing browser state changes in real-time.



Automation Audit Reports Table

Displays the details of completed and verified PO runs with execution times, statuses, and exceptions, allowing admins to view results and download full artifact zip logs.

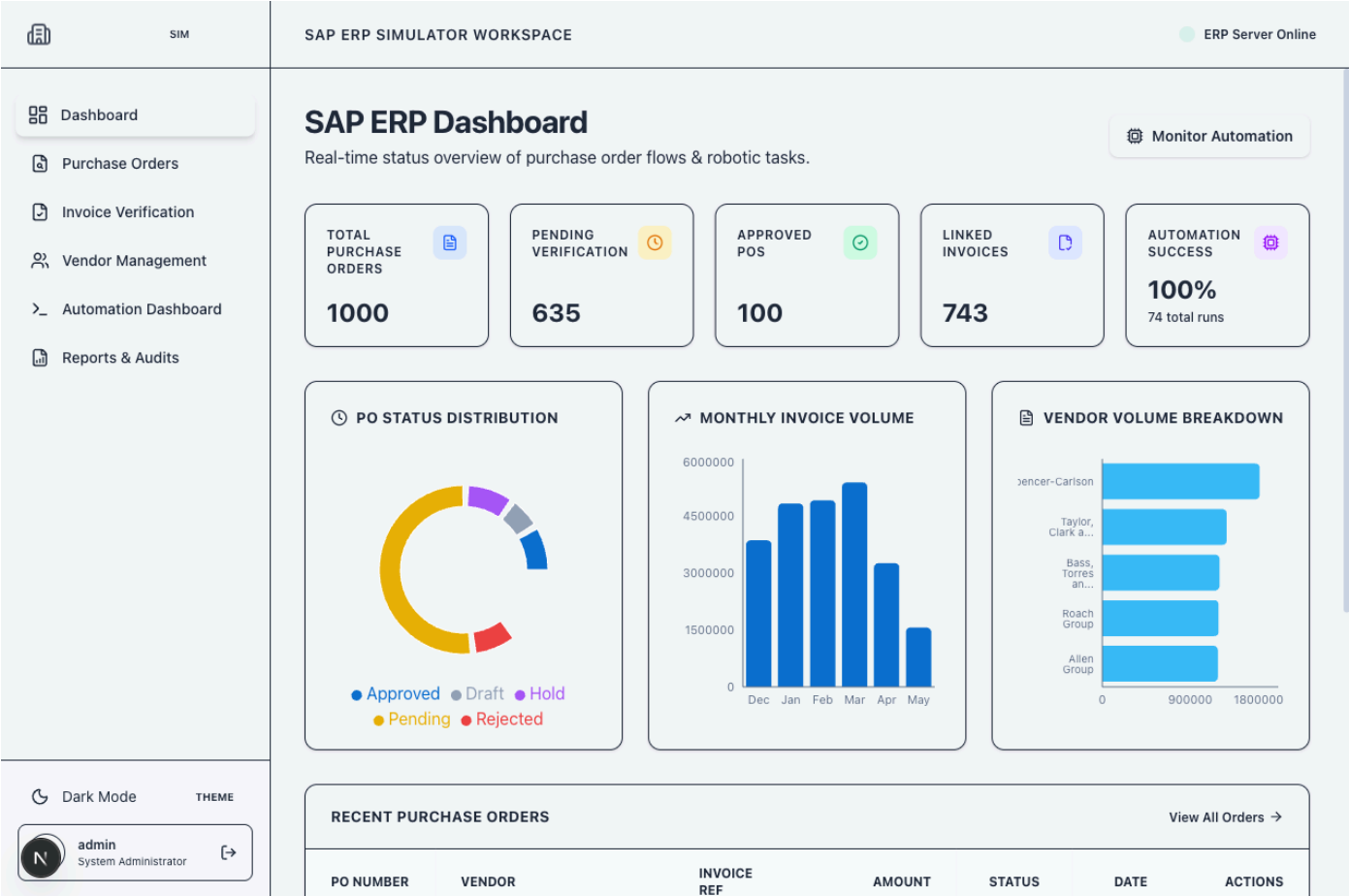
	SAP ERP SIMULATOR WORKSPACE						ERP Server Online
	1. LOAD TARGET RECORDS (EXCEL)						
	2. LAUNCH EXECUTABLE ACTION						
	Session Status: RUNNING						
	Processing Record: 45000363						
	Queue Progress (2/50) 4%						
	SUCCESS 2						
	FAILED 0						
	LIVE BROWSER VIEW: PO 45000363						
	Dashboard Screen						
	PO Search Table						
	PO Details Card						
	Invoice Tab						
	Approval Section						
	AUTOMATION AUDIT REPORTS						Download All Screenshots (.ZIP)
	PO NUMBER	VENDOR	INVOICE	STATUS	EXECUTION TIME	FINISHED AT	DETAILS
	45000109	Hickman LLC	INV-GEN-70852	Success	10.06s	2026-05-20 00:21:22	Completed without exceptions.
	45000187	Colon, Blankenship and Turner	INV-GEN-33461	Success	10.07s	2026-05-20 00:21:32	Completed without exceptions.
	45000203	Aguirre-Lee	INV-GEN-52182	Success	10.05s	2026-05-20 00:21:52	Completed without exceptions.
	45000223	Vaughan, Rodriguez and Smith	INV-900223	Success	10.08s	2026-05-20 00:21:42	Completed without exceptions.
	45000265	Spencer-Carlson	INV-900265	Success	10.09s	2026-05-20 00:20:42	Completed without exceptions.

Automation Checkpoints (Visual Evidence)

Below are the actual screenshots captured by the Playwright engine at each checkpoint during the automated execution:

Checkpoint 1: Portal Dashboard Page (dashboard.png)

Captures the initial workspace entry and metric summaries before the run starts.



Checkpoint 2: PO Search Results Grid (search-results.png)

Captured immediately after entering the target PO number and applying the search filter.

SIM

Dashboard

Purchase Orders

Invoice Verification

Vendor Management

Automation Dashboard

Reports & Audits

SAP ERP SIMULATOR WORKSPACE

ERP Server Online

Purchase Order Workspace

Search, verify, and action enterprise purchase orders and invoices.

PO NUMBER

VENDOR

All Vendors

STATUS

All Statuses

SORT BY

PO Number (Desc)

SCOPE:

☒ All Orders

☐ Active Only

☐ High Value (>\$50K)

Reset

Search

SEARCH RESULTS (1 FOUND)

PO NUMBER	VENDOR	INVOICE NUMBER	AMOUNT	STATUS	CREATED DATE	ACTIONS
45000082	Luna-Fuentes	-	144,323.35 GBP	Approved	2026-01-07	View Details

Dark Mode

THEME

admin

System Administrator

Checkpoint 3: Interactive Verification Workspace (po-details.png)

Captures the material items table and the robot filling out the checkboxes and remarks.

SIM

Dashboard

Purchase Orders

Invoice Verification

Vendor Management

Automation Dashboard

Reports & Audits

Dark Mode

THEME

N

admin

System Administrator

SAP ERP SIMULATOR WORKSPACE

ERP Server Online

LINE ITEM BREAKDOWN

ITEM CODE	DESCRIPTION	QTY	UNIT PRICE	TAX %	TOTAL
MAT-001	Enterprise Server Rack Installation	1	72161.68	18%	72161.68
MAT-042	High Performance Switchboards	15	2886.47	18%	43297.00
MAT-089	Fiber Optic Cabling Bundle	100	288.65	18%	28864.67

INTERACTIVE VERIFICATION FORM

Three-Way Matching Status:
☒ Verified Match (PO/GR/IR) ☐ Mismatch Found

Verification Checklist:
☒ Verify quantities & item codes ☒ Verify unit prices & tax charges

Verification Remarks
Automated Playwright verification run. expected_status: Approved, invoice: INV-GEN-59005

NH 99387

CONTACT EMAIL
info@luna-fuentes.com

PHONE
001-397-999-5527

CONTRACT TERMS
Net 30

WORKFLOW STATUS ACTIONS

Approve, reject, or put this purchase order on hold. Taking action will notify the vendor and release payments.

✓ Approve Order

✗ Reject Order

🛑 Place On Hold

Checkpoint 4: Linked Invoice Scan Details (invoice-tab.png)

Captures the active invoice scanning card data extracted using Ollama VLM OCR.

SIM

SAP ERP SIMULATOR WORKSPACE

ERP Server Online

Dashboard

Purchase Orders

Invoice Verification

Vendor Management

Automation Dashboard

Reports & Audits

Dark Mode

THEME

admin

System Administrator

144,323.35

2026-01-07

Approved

VEND-100044

GBP

Items & Verification

Invoice Details

Attachments

Audit Trail

Payment Terms

No Invoice Linked

This PO does not have an active invoice linked to it. Run automation to map invoice scans.

COMPANY NAME

Luna-Fuentes

TAX ID / GST NUMBER

64AAAAA7735A8Z1

ADDRESS

12432 Kathleen Shoal, East Stephaniefort, NH 99387

CONTACT EMAIL

info@luna-fuentes.com

PHONE

001-397-999-5527

CONTRACT TERMS

Net 30

WORKFLOW STATUS ACTIONS

Approve, reject, or put this purchase order on hold. Taking action will notify the vendor and release payments.

✓ Approve Order

✗ Reject Order

🔄 Place On Hold

Checkpoint 5: Approval Execution Confirmation (approval-section.png)

Captured after the robot clicks "Approve Order", confirming the backend updated state successfully.

- **Terminal 1 (SAP Backend on Port 8000):**

```
source .venv/bin/activate  
python3 apps/sap-simulator/backend/main.py
```

- **Terminal 2 (Automation API on Port 8001):**

```
source .venv/bin/activate  
python3 apps/automation-engine/api/main.py
```

4. Launch Frontend Next.js Server:

- **Terminal 3 (Next.js on Port 3000):**

```
cd apps/sap-simulator/frontend  
npm install  
npm run dev
```

Option B: Docker Compose Running (Headless Mode)

To run the entire system containerized:

```
docker-compose up --build
```

Note: Playwright will run in headless mode inside the container.

Credentials & Testing

- **Login URL:** <http://localhost:3000>
- **Username:** [admin](#)
- **Password:** [admin123](#)

- **Demo File:** Click the link in the RPA Dashboard to download the [sample_po_list.xlsx](#), then drag-and-drop it into the uploader to trigger the run!